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PRESENCE OF FOLLICLE STIMULATING HORMONE RECEPTORS IN TESTIS AND EPIDIDYMIS OF A LACERTILIAN SPECIES, *UROMASTIX HARDWICKII*

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INTRODUCTION

Regulation of gonadal function by pituitary hormones has been extensively studied in mammalian species (Russel *et al.,* 1987; Sharp, 1993) but remains controversial in reptilian species (Callard *et al.,* 1978, Licht, 1984; Arslan *et al.,* 1977, 1986a, b; Qureshi, *et al.,* 1992). Dependence of gametogenic and steroidogenic activity in reptilian gonads on a single FSH-like molecule (Licht and Stockwell-Hartree, 1971; See; Fox, 1977) was a clear deviation from the tetrapod pattern of having two distinct gonadotropins, FSH and LH, regulating gametogenic and steroidogenic activity, respectively. The nature and precise physiological function of FSH and LH on gonads and reproductive tract still remains obscure (Licht, 1984; Qureshi *et al.,* 1992). However, distinct molecular species in certain lower vertebrates have also been identified (Donaldson *et al.,* 1972, Licht, 1984). Likewise, structural and functional integrity of epididymis was shown to be androgen dependent in a reptilian species (Dufaure *et al.,* 1986; Faure *et al.,* 1990, 1991) and the suggestion was, however, made for the involvement of non androgenic factors in mRNA distribution in the duct (Faure *et al.,* 1991) supporting the observations of FSH like molecules playing a major role in the regulation of duct function in two reptilian species, *Uromastix hardwickii* (Arslan *et al.,* 1986a, b; Qureshi *et al.,* 1988, 1992) and *H. flaviviridis* (Haider and Rai, 1987). Binding of peptide hormones to the target cell membrane is the first step in the mechanism of action of gonadotropin hormones extensively studied and characterized mostly in the laboratory animals both *in vivo* and *in vitro* (Dias, 1992a, b; Hsueh and Lapolt, 1992; Huhtaniemi *et al.,* 1992; Griswold, 1993; Liu *et al.,* 1994; Tena-Sempere *et al.,* 1994). Surface receptors for FSH (Bhalla and Reichert, 1974; Berman and Sairam, 1984; Reichert and Dattatreyamurty, 1989) and LH (Dufau and Catt, 1978; Ascoli and Segaloff, 1989; McFarland *et al.,* 1989; Segaloff and Ascoli, 1993) have been demonstrated in testicular tissue from many species and show similarity in many aspects (Davies *et al.,* 1979; Berman and Sairam, 1984) but little information is available for these receptors in lower vertebrates. LH receptors and not FSH receptors are present on leydig cells whereas FSH receptors are present in the seminiferous tubules on sertoli cells and possibly on spermatogonia (Dias, 1992a). Thus, in view of controversial status of FSH and LH on regulation of reptilian testes and epididymis experiments were conducted to identify the binding sites for h- FSH in preference to h-LH in the testes and epididymis obtained from a reptilian species *Uromastix hardwickii* during its active phase of annual reproductive cycle.

MATERIALS AND METHODS

Preparation of tissue fractions:

The animals were dissected out and testes, epididymis, heart, liver, muscle and brain tissues were removed. Tissue fractions were prepared as enriched membrane preparations obtained from fully mature testes (from months of March and April) essentially as previously described (LeGac *et al.*, 1988). Briefly, the freshly dissected testes were decapsulated (at 4°C), minced with scissor, rinsed and suspended in 25 mM Tris-HCl buffer (pH 7.2) containing 0.1 M sucrose (2.5 ml/g tissue). Pooled testes were homogenized by hand held glass tissue grinder using 8 strokes (up & down). Homogenates from epididymis, cleared from its surrounding tissue, liver, muscle and brain were prepared using the same procedure, while being kept over ice. The homogenates were filtered through 4 layers of cheese cloth and the filtrates were subjected to centrifugation first at low speed (600 x g) for 30 minutes to eliminate cellular debris, including undisturbed tissue, broken cells nuclei and most of spermatocyte, and then at 8000 x g for 30 minutes to 1 hour at 4°C and the resulting pellet was washed and resuspended in assay buffer, Tris-HCl, pH 7.2 (1g/2ml) containing 10mM $MgCl_2$ and 0.3% BSA and stored at - 20°C till assayed.

Protein estimation:

Protein content was determined by Lowry's method (1951) using bovine serum albumin as standard.

Labelled Hormones:

The ^{125}I labelled hormones, h-FSH (Batch No.F-9012), h-LH (L 9012) and h-PRL (P 9021) were a gift rom WHO (specific activities were in the range of 50-80 Ci/ug).

Binding of ^{125}I -labelled hormones to tissue fractions:

The assays of binding of ^{125}I-labelled hormones to each of the tissue fractions were performed in triplicate in 10 x 75 mm disposable polystyrene tubes containing 50000 cpm (~ 400 pg), ^{125}I-labelled hormone, 100-300 ug% protein equivalent of testicular fractions and 50 ul unlabelled hormone and/or assay buffer to constitute a final volume of 250 ul per tube. The assay buffer consisted of 25 mM Tris-HCl, pH 7.2 containing 0.1% BSA and 10 mM $MgCl_2$. The tubes were mixed using vortex and incubated in a continuously shaking Dubnoff water bath (70 oscill./min) for 3 hours at 30°C. The reaction was terminated by the addition of 100 ul chilled assay buffer followed by centrifugation at 2900 g for 15 min. at 4°C and the pallets washed twice with cold buffer. The supernatant was removed by aspiration, the inverted tubes left on filter paper for a few minutes till pellet seemed devoid of buffer. The pellet associated radioactivity was counted in Packard Gamma counter (75% efficiency) in RIA lab of NRIFC. Non- specific binding was determined in the presence of excess of respective unlabelled hormone. The difference between the total radioactivity bound and non-specific binding was defined as the amount specifically bound and expressed as percentage of the total counts put into the tubes.

Statistical Analysis:

The data were analyzed for significance by student's 't' test.

RESULTS AND DISCUSSION

The tissue homogenates containing equal amounts from testes, epididymis, muscle, liver and brain were individually incubated with ^{125}Ih-FSH and ^{125}Ih-LH to find the tissue and hormone specificity. It was found that maximum specific binding (Fig. 1) was obtained in testes (20-25%) followed by that in epididymis (12-15%) for h-FSH. Binding for both hormones was almost negligible in non target tissues as liver and muscle. In parallel experiments binding of both testes and epididymis was comparatively very low for h-LH.

In another set of experiment the increasing concentration of tissue homogenates of both testis and epididymis gradually increased the specific binding with h-FSH. On the contrary, h-LH failed to show any such relationship with epididymis although testicular homogenate resulted more or less as in case of h-FSH. Muscle and brain tissues did not produce such linear relationship.

Further, on increasing the concentration of unlabelled h-FSH in the incubation media, the ^{125}Ih-FHS binding to these tissues was found to be inhibited (Fig. 2). Thus, the present study demonstrates that ^{125}Ih-FSH specifically binds to both testicular and epididymal tissues in a reptilian species during the active phase of the annual reproductive cycle. *In vitro* specific binding of ^{125}Ih-FSH to testicular tissue of uromastix has previously been found to be maximum during the reproductive phase (Jalali *et al.*, 1984). The binding appears to be tissue specific for ^{125}Ih-FSH and are in line with the finding of the proposed physiological role of FSH in the testicular and epididymal function of reptiles (Arslan *et al.*, 1986a, b; Qureshi *et al.*, 1992). PMSG with predominant FSH like activity was found to increase both steroidogenic and spermatogenic activity in this species (Jalali *et al.*, 1975, 1979). It may have be pointed out that the tissue crude homogenate is primarily the leyding cell deficient and it is presumed that the ligand binds specifically to the sertoli cells. However, binding to leydig cells can not be ruled out. These experiments were carried out at 30°C and it has been found previously that hormone binding of gonadotropin is obtained maximally at this temperature (Licht and Midgley, 1976) in lizard, snake and turtle, although time dependent binding pattern for different species of reptiles has also been documented (3-8 hrs). In the present study the incubations was done for the same time in all experiments. The percentage of radiological specific binding is comparatively low as compared to those obtained from a variety of mammalian gonadotropin receptor system using ^{125}Ih- FSH. Experiments showing inhibition of binding by unlabelled FSH documented sensitivity of both testicular and epididymal tissues as derived from studies in mammalian system (Means and Vitukaitis, 1972; Bhalla and Richert, 1974).

The specific binding of ^{125}Ih-FSh and not of ^{125}Ih-LH to homogenates of reptilian epididymis supports the suggestion of the role of FSH in the epididymal physiology when FSH was found to a stimulate the epididymal principal cells (Arslan *et al.*, 1986a, b; Qureshi *et al.*, 1988) and their protein profiles (Qureshi *et al.*, 1992). Previously, relative effects of mammalian FSH and LH on the regressed epididymis in wall lizard, *H. flaviviridis* (Haider and Rai, 1987) revealed that only FSH was capable of stimulating the growth and secretory activity of the duct and FSH was shown to maintain the sexual segment in the same species (Reddy and Prasad, See: Fox, 1977). Three mammalian gonadotropins (FSH, HCG and PMSG) were found to produce hypertrophy of leydig cells and a significant increase in the epithelial cells of quiescent epididymis (Jones, 1973). Potent steroidogenic activity of O-FSH in male Anolis (Licht *et al.*, 1974, 1979) and of bovine FSH in sea turtle (Owens *et al.*, 1976) were confirmed in turtles (Callard *et al.*, 1976).

In hypophysectomized reptiles mammalian FSH alone can maintain gametogenesis in both sexes were LH is not effective (Licht, 1984). During the non breeding season, administration of FSH and PMSG to hypophysectomized lizard could initiate spermatogenesis leading to formation of spermatozoa and the development of sexual segment, whereas LH and HCG had no effect of (Reddy and Prasad, 1970). Similarly FSH or PMSG alone could stimulate regressed testis in *Uromastix hardwickii* (Jalali *et al.*, 1975). Two distinct gonadotropins in pituitary of turtles and alligators immunologically related to the mammalian FSH and LH (Licht *et al.*, 1977, 1979) have been isolated (Licht and Papkoff, 1978). In the snake however, all the evidences seem to indicate the existence of a single gonadotropin not resembling either FSH or LH of mammals (See: Licht, 1984). This feature of reptiles is therefore, a major evolutionary divergence among tetrapods. The scanty information only in some turtles (Licht 1979) and in Cobra (Bonna-Gallo *et al.*, 1980) show a clear seasonal pattern of secretion of the gonadotropin as well as the response of the gonads to these hormones.

FSH is required for normal reproductive function in particular for the initiation and maintenance of spermatogenesis (Griswold, 1993). FSH belonging to the glycoprotein family and its biochemical actions are expressed through its interaction with cell surface FSH receptors (FSHR). Considerable progress has already been made in understanding the structure and function of its ligand FSH (Dias, 1992a; Keutmann, 1992). The receptor molecules exert their major effects by activation of adenyl cyclase after hormone binding (Reichert and Datta Treyamurty, 1989). The LH receptor (LHR) like other members of the G-protein-coupled receptor superfamily including FSHR contains seven putative transmembrane domains. It is expressed in the gonads and targets the action of LH to leydig cell in the testis and to thecal, luteal and granulosa cells in the ovary and its gene is large (70 Kb) encoded by an 11-exon/10-intron single copy gene (McFarland *et al.*, 1989; Hsueh and Lapolt, 1992; Segaloff and Ascoli, 1993).

Cloning of cDNA's and isolation of genes for all gonadotropin hormone receptors have contributed to a better understanding of the structure of glycoprotein hormone receptors (Ascoli and Segaloff, 1989; Dias *et al.*, 1992a, b) and primary sequence of FSHR has recently been found to be homologous to LHR and TSHR (Liu *et al.*, 1994). The primary sequence of FSHR has been determined by recent cloning techniques and the complimentary DNA (cDNA) sequence from rats (Sprengel *et al.*, 1992) and human (Kelton *et al.*, 1992, Tilly *et al.*, 1992) has been achieved.

Temperature and photoperiods significantly affect the gonadal activity in reptiles (Licht, 1984). In seasonal breeding mammals, short photoperiod induced changes are associated with decreased FSH, LH and prolactin (Yellon & Goddman, 1987) and its receptor content of the testes (Bex and Bartke, 1978; Klemecke *et al.*, 1987; Bartke and Stegar, 1992). Importance of FSH for its critical role in androgen synthesis i.e. enhancing the efficacy of LH action testicular steroidogenesis has been suggested (Grimek *et al.*, 1976; Chandrasekhar *et al.*, 1994). Additionally, PRL rather than gonadotropin, is known to regulate testicular LH receptors in golden hamster (Bex *et al.*, 1978; Bartke *et al.*, 1987; Barkte and Stegar, 1992). Cell-Cell communication and interaction with the testes via paracrine and Juxtracrine mechanisms are thought to be responsible for local control of spermatogenesis in mammalian species (Russel *et al.*, 1987; Sharp, 1993). Sertoli cells have been shown to secrete a variety of proteins that directly or indirectly support the growth, differentiation and development of germ cells and conversely, germ cells and their derived proteins have been shown to modulate secretory function of sertoli cells (Djakiew and Dym, 1988; Onoda and Djakiew, 1990, 1991). Such an approach to study the reptilian gonadal regulation has yet to be undertaken.

The present study does support the suggestion that FSH like molecules to have directly on indirectly a prominent role in the testicular and epididymal regulation of this seasonally breeding reptilian species. Experiments are needed to characterize the pituitary gonadotropin receptors, their evolution during annual reproductive cycle employing the tools of molecular biology as currently being performed for mammalian receptor system (Liu *et al.*, 1994; Reichert, 1994; Tena-Sempere *et al.*, 1994, Valove *et al.*, 1994).

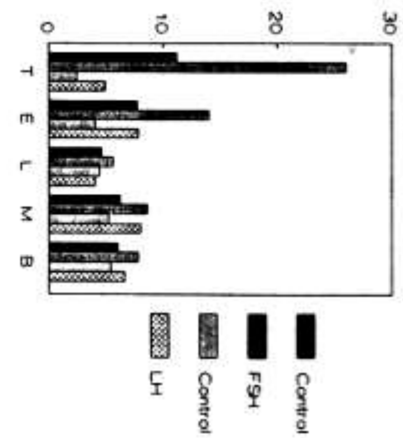

Fig.I: Specific binding (%) of ^{125}I h-FSH and ^{125}I hLH to membrane preparations from lizard test is, epididymis, liver, muscle and brain.

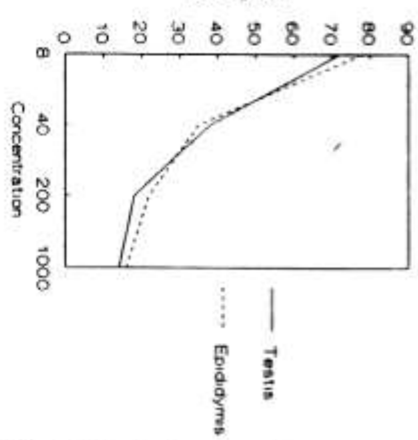

Fig.II: Inhibition of ^{125}I h-FSH relative specific binding to testicular (------) and epididymal (- - - -) homogenates concentration of unlabelled O-FSH.

REFERENCES

Arslan M., Lobo J., Zaidi A.A. and Qazi M.H. (1977). *Gen. Comp. Endo.* 33: 160.

Arslan M., Jalali S. and Qazi M.H. (1986a). *Pak. J. Zoology. 18(4): 371.*

Arslan M., Jalali S., Nasreen I. and Qazi M.H. (1986b). *Zool. Biol.* 5: 251.

Ascoli M. and Segaloff D.L. (1989). *Endocr. Rev.* 10: 27.

Bartke A. and Steger R.W. (1992). *Proc. Soc. Exp. Biol. Med.* 199: 139.

Bartke A., Amador A. Chadrashekar V. and Klemcke H.G. (1987). *J. Steroid Biochem.* 27: 581.

Berman M.I. and Sairam M.R. (1984). *J. Reprod. Fert.* 70: 463.

Bex F.J. and Bartke A. (1978). *Endocrinology.* 100: 1223.

Bex F.J., Bartke A., Goldman G.D. and Dalterio S. (1978). *Endocrinology.* 103: 2069.

Bhalla V.K. and Reichert L.E. Jr. (1974). *J. Biol. Chem.* 249: 43.

Bonna-Gallo A., Licht P., Mackenzie D.S. and Lofts B. (1980). *Gen. Com. Endo.* 42: 477.

Callard I.P., Callard G.V., Lance V. and Eceles S. (1976). *Gen. Comp. Endo.* 30: 347.

Callard I.P., Callard G.V., Lance V., Bolaffi J.L. and Roset J.A. (1978). *Biol. Reprod.* 18: 16.

Chandrashekar V., Majumdar S.S. and Bartke A. (1994). *Biol. Rep.* 50: 82.

Davies T.F., Walsh P.C., Hodgen G.D., Dafau M.L. and Catt K.J. (1979). *J. Clin. Endocr. Metab.* 48: 680.

Dias J.A. (1992a). *Trends Endo. Metab.* 3: 24.

Dias J.A. (1992b). *Biochim. Biophys. Acta.* 1135: 278.

Djakiew D. and Dym M. (1988). *Biol. Reprod.* 39: 1193.

Donaldson E.M., Yamazaki F., Dye H.H. and Philleo W.W. (1972). *Gen. Comp. Endo.* 18: 469.

Dufau M.I. and Catt K.J. (1978). *Vitams Horm.* 36: 461.

Dufaure J.P., Courty Y., Depeiges A., Mesure M. and Chevalier M. (1986). *Biol. Reprod.* 35: 667.

Faure J., Chevalier M., Mesure M. and Dufaure J.P. (1990). *J. Histochem. Cytochem.* 38(4): 533.

Faure J., Ghyselinck N.B., Jmenez C. and Dufaure J.P. (1991). *Biol. Reprod.* 44: 13.

Fox H. (1977). The urinogenital system. In: Biology of reptili, Morphology Ed. Gans C. and Parison I. Acad. Press Vol.6, p.1.

Grimek H.H., Nuti L.S., Nuti K.M. and McShan W.H. (1976). *Endocrinology,* 98: 105.

Griswold M.D. (1993). Actions of FSH on mammalan Sertoli cells. In: Russell L.D. Griswold M.D. (Eds). The Sertoli Cell. Cache River Press, Clearwater, p.494.

Haider S. and Rai U. (1987). *J. Morphology.* 191: 151.

Hsueh A.J.W. and LaPolt P.S. (1992). *Trends Endocrinol Metab.* 3: 164.

Huhtaniemi I.T., Eskola V., Pakarnen P., Matikanen T. and Sprengel R. (1992). *Mol. Cell. Endo.* 88: 55.

Jalali S., Arslan M. and Qazi M.H. (1975). *Isl. J. Sci.* 2: 10.

Jalali S., Aslam N. and Arslan M. (1984). *Biologia.* 30(2): 297.

Jalali S., Arslan M., Qureshi S. and Qazi M.H. (1979). *Gen. Comp. Endo.* 30: 162.

Jones R.E. (1973). *Comp. Endo.* 20: 567.

Kelton C.A., Cheng S.V.Y., Nugent N.P., Schweichkardt R.L., Rosenthal J.L., Overton S.A., Wands G.D., Kuzeja J.B., Luchette C.A. and Chappel S.C. (1992). *Mol. Cell. Endo.* 89: 141.

Keutmann H.T. (1992). *Mol. Cell Endocrinology.* 86: C1-C6.

Klemcke H.G., VanSickle M., Barke A., Amador A. and Chandrashekar V. (1987). *Biol. Reprod.* 37: 356.

Le-Gac F. and Breton B. Bougoussa M. (1988). *Fish Physiol Biochem.* 5: 209.

Licht P. (1979). *Annual Review of Physiology.* 41: 337.

Licht P. (1984). Reptiles. In: Marshall's Physiology of Reproduction. Vol.1: Reproductive Cycles of Vertebrates (Eds) Lamming G.E. Churchill Livingstone pp.206.

Licht P. and Midgley A.R. Jr. (1976). *Biol. Reprod.* 15: 195.

Licht P. and Papkoff H. (1978). Species specificity as a key to structural evolution in pituitary glycoprotein hormones. In: Gaillard, P.J., Boer H.H. (Eds.) Comparative Endocrinology. Elsevier./North-Holland, Amsterdam.

Licht P. and Stockwell Hartree A. (1971). *J. Endocrinol.* 49: 113.

Licht P., Papkoff H., Farmer S.W., Muller C.H., Tsui H.W. and Crews D. (1977). *Recent Progress in Hormone Research.* 3: 169.

Licht P., Papkoff H., Goldman B.D., Follett B.K. and Scanes C.G. (1974). *Gen. Comp. Endo.* 24: 168.

Liu X., DePasquale J.A., Griswold M.D. and Dias J.A. (1994). *Endocrinology.* 135(2): 682.

Lowry O.H., Resebrough N.J., Farr A.L. and Randal P.J. (1951). *J. Biol. Chem.* 193: 265.

McFarland K.C., Sprengel R., Phillips H.S., Kohler M., Rosemblit N., Nikolics K., Segaloff D.L. and Seeburg P.H. (1989). *Science.* 245: 494.

Means A.R. and Vaitukaitis J. (1972). *Endocrinology.* 90: 39.

Onoda M. and Djakiew D. (1991). *In Vitro Cell Dev Biol.* 27A: 215.

Onoda M. and Djakiew D. (1990). *Mol. Cell Endocrinol.* 73: 35.

Qureshi M.A., Shaheen A. and Qureshi N.M. (1988). Hormonal regulation of epididymal function. *8th Pak. Congr. Zool.* p.83.

Qureshi M.A., Fayyaz A. and Qureshi N.M. (1992). Effect of gonadotropins and steroids on the epididymal proteins during inactive phase of reproductive cycle of lacertilian species. International Sym. on Epididymis and male fertility, Hong Kong. pp.

Reddy P.R.K. and Prasad M.R.N. (1970). *Ruppel. Gen. Comp. Endo.* 14: 15.

Reichert L.E. (1994). *Mol. Cell. Endo.* 100: 21.

Reichert L.E. and Dattatreyamurty B.D. (1989). *Biol. Reprod.* 40: 13.

Russel D.L., Alger L.E. and Nequin L.G. (1987). *Endoc.* 120(4): 1615.

Segaloff D.L. and Ascoli M. (1993). *Endocr. Rev.* 14: 324.

Sharpe R.M. (1993). Experimental evidence for Sertoli-germ cell and Sertoli-Leydig cell interactions. In: Russell L.D., Griswold M.D. (Eds). The Sertoli Cell. Clearwater F.L., Cache River Press, p. 391.

Sprengel R., Braun T., NikolicsK., Segaloff D.L. and Seeburg P.H. (1990). *Mol. Endo.* 4: 525.

Tena-Sempere M., Zhang Fu-Ping and Huhtaniemi I. (1994). *Endocrinology* 135(3): 1018.

Tilly J.L., Aihara T., Hishimori K., Jia S.C., Billig H., Kowalski K.I., Perlas E.A. and Hsueh A.J.W. (1992). *Endo.* 131: 799.

Tsui H.W. (1976). *Gen. Comp. Endo.* 28: 386.

Valove F.M., Finch C., Anasti J.N., Froehlich J. and Flack M.R. (1994). *Biol. Reprod.* 135(6): 2657.

Vellon S.M. and Goldman B.D. (1987). *J. Reprod. Fertil.* 80: 167.